Fourier Series and PDEs Problem Sheet 2

1. (a) Establish the *orthogonality relations* given by

$$\begin{aligned}\int_{-\pi}^{\pi} \cos(mx)\cos(nx)\,\mathrm{d}x &= \pi\delta_{mn},\\ \int_{-\pi}^{\pi} \sin(mx)\cos(nx)\,\mathrm{d}x &= 0,\\ \int_{-\pi}^{\pi} \sin(mx)\sin(nx)\,\mathrm{d}x &= \pi\delta_{mn},\end{aligned}$$

where m and n are positive integers and δ_{mn} is Kronecker's delta.

[*You may find it helpful to use the identities* $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$.]

(b) Let f be a continuous periodic function of period 2π. Suppose that f may be expressed in the form of the Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \big(a_n \cos(nx) + b_n \sin(nx)\big).$$

Assuming that the orders of summation and integration may be interchanged, derive

(i) integral expressions over $[-\pi, \pi]$ for the Fourier coefficients a_n and b_n;

(ii) *Parseval's identity* in the form

$$\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)^2\,\mathrm{d}x = \frac{1}{2}a_0^2 + \sum_{n=1}^{\infty}\left(a_n^2 + b_n^2\right).$$

[*For (ii) you may find it helpful to integrate the product of* f *and the Fourier series for* f.]

2. The periodic function f of period 2π is defined by

$$f(x) = x\sin(px) \quad \text{for } -\pi < x \le \pi,$$

where p is a positive integer.

(a) Is f even or odd? Use integration by parts to show that if m is a non-zero integer then

$$\int_0^{\pi} x\sin(mx)\,\mathrm{d}x = \frac{\pi(-1)^{m+1}}{m}.$$

Hence, or otherwise, show that the Fourier series for f is given by

$$f(x) \sim \frac{(-1)^{p+1}}{p} - \frac{\cos(px)}{2p} + \sum_{\substack{n=1\\ n\neq p}}^{\infty} \frac{2p(-1)^{n+p}}{n^2-p^2}\cos(nx).$$

[*You may find it helpful to use the identity* $2\sin A\cos B = \sin(A+B) + \sin(A-B)$.]

(b) Use the Fourier convergence theorem to deduce that

$$\sum_{n=2}^{\infty}\frac{(-1)^n}{n^2-1} = \frac{1}{4} \qquad \text{and} \qquad \sum_{n=3}^{\infty}\frac{1}{n^2-4} = \frac{25}{48}.$$

(c) Given that Parseval's identity holds for f, evaluate the sum

$$\sum_{\substack{n=1\\ n\neq p}}^{\infty}\frac{1}{(n^2-p^2)^2}.$$

3. The even periodic function f of period 2π is defined by

$$f(x) = \frac{x^3}{6} - \frac{\pi x^2}{4} + \frac{\pi^3}{24} \quad \text{for } 0 \leq x \leqslant \pi.$$

(a) Determine $f'(x)$ and $f''(x)$ for $x \in \mathbb{R}$. Sketch on separate axes the graphs of $f''(x)$, $f'(x)$ and $f(x)$ for $x \in (-2\pi, 2\pi]$.

(b) Use integration by parts to show that if n is a positive integer then

$$\int_0^\pi f(x)\cos(nx)\,\mathrm{d}x = -\frac{1}{n}\int_0^\pi f'(x)\sin(nx)\,\mathrm{d}x = -\frac{1}{n^2}\int_0^\pi f''(x)\cos(nx)\,\mathrm{d}x.$$

(c) Show that the Fourier series for f'' is given by

$$f''(x) \sim -\frac{4}{\pi}\sum_{m=0}^{\infty}\frac{\cos\big((2m+1)x\big)}{(2m+1)^2}$$

and find the Fourier series for f' and f. What is the link between the smoothness of each function and the rate of convergence of its Fourier series?